Now $z = \left(\frac{19+9i}{5-3i}\right)^{15} - \left(\frac{8+i}{1+2i}\right)^{15}$

$\Rightarrow z = (2+3i)^{15} - (2-3i)^{15}$. (by (1) and (2))

Then by definition, $\overline{z} = \left(\overline{(2+3i)^{15} - (2-3i)^{15}}\right)$

$= \left(\overline{2+3i}\right)^{15} - \left(\overline{2-3i}\right)^{15}$ (using properties of conjugates)

$= (2-3i)^{15} - (2+3i)^{15} = -\left((2+3i)^{15} - (2-3i)^{15}\right)$

$\Rightarrow \overline{z} = -z$.

Therefore, $z = \left(\frac{19+9i}{5-3i}\right)^{15} - \left(\frac{8+i}{1+2i}\right)^{15}$ is purely imaginary.

EXERCISE 2.4

1. Write the following in the rectangular form:
   (i) $\overline{(5+9i)+(2-4i)}$ (ii) $\frac{10-5i}{6+2i}$ (iii) $\overline{3i} + \frac{1}{2-i}$
2. If $z = x + iy$, find the following in rectangular form.
   (i) $\mathrm{Re}\left(\frac{1}{z}\right)$ (ii) $\mathrm{Re}(i\overline{z})$ (iii) $\mathrm{Im}(3z + 4\overline{z} - 4i)$
3. If $z_1 = 2 - i$ and $z_2 = -4 + 3i$, find the inverse of $z_1 z_2$ and $\frac{z_1}{z_2}$.
4. The complex numbers u, v, and w are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$.
   If $v = 3 - 4i$ and $w = 4 + 3i$, find u in rectangular form.
5. Prove the following properties:
   (i) z is real if and only if $z = \overline{z}$ (ii) $\mathrm{Re}(z) = \frac{z+\overline{z}}{2}$ and $\mathrm{Im}(z) = \frac{z-\overline{z}}{2i}$
6. Find the least value of the positive integer n for which $\left(\sqrt{3}+i\right)^n$
   (i) real (ii) purely imaginary.
7. Show that (i) $\left(2+i\sqrt{3}\right)^{10} - \left(2-i\sqrt{3}\right)^{10}$ is purely imaginary
   (ii) $\left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real.

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